

CONSTITUTION

LLMProvider — Constitution

Inherits from: HelixConstitution/Constitution.md — all universal clauses in that document apply unconditionally. Rules below extend or tighten them; they **MUST NOT** weaken any inherited clause.

Status: Active. This document is the project's authoritative rule set. When a rule here conflicts with CLAUDE.md, AGENTS.md, or any guide, the Constitution wins.

Mission

See README.md.

Mandatory Standards

1. **Reproducibility:** every change is reproducible from a clean clone (`git clone <repo> && <project bootstrap>`); no hidden steps.
2. **Tests track behavior, not code:** test what the user-visible behavior is, not what the implementation looks like.
3. **No silent skips, no silent mocks above unit tests.**
4. **Conventional Commits** for all commits.
5. **SSH-only for git operations** (`git@...`); HTTPS prohibited.

Numbered Rules

CONST-033 — Host Power Management is Forbidden

Status: Mandatory. Non-negotiable. Applies to every project, submodule, container entry point, build script, test, challenge, and systemd unit shipped from this repository.

Rule: No code in this repository may invoke a host-level power-state transition (suspend, hibernate, hybrid-sleep, suspend-then-hibernate, poweroff, halt, reboot, kexec) on the host machine. This includes — but is not limited to:

- `systemctl {suspend,hibernate,hybrid-sleep,suspend-then-hibernate,poweroff,halt,reboot,kexec}`
- `loginctl {suspend,hibernate,hybrid-sleep,suspend-then-hibernate,poweroff,halt,reboot}`
- `pm-{suspend,hibernate,suspend-hybrid}`
- `shutdown {-h,-r,-P,-H,now,--halt,--poweroff,--reboot}`
- Dbus calls to `org.freedesktop.login1.Manager`.
`{Suspend,Hibernate,HybridSleep,SuspendThenHibernate,PowerOff,Reboot}`
- Dbus calls to `org.freedesktop.UPower`.
`{Suspend,Hibernate,HybridSleep}`
- `gsettings set ... sleep-inactive-{ac,battery}-type` to any value other than 'nothing' or 'blank'

Why: The host runs mission-critical parallel CLI-agent and container workloads. On 2026-04-26 18:23:43 the host was auto-suspended by the GDM greeter's idle policy mid-session, killing HelixAgent and 41 dependent services. Recurring memory-pressure SIGKILLS of `user@1000.service` (perceived as "logged out") have the same outcome. Auto-suspend, hibernate, and any power-state transition are unsafe for this host.

Defence in depth (mandatory artifacts in every project):

1. `scripts/host-power-management/install-host-suspend-guard.sh` — privileged installer, manual prereq, run once per host with `sudo`. Masks `sleep.target`, `suspend.target`, `hibernate.target`, `hybrid-sleep.target`; writes `AllowSuspend=no` drop-in; sets `logind IdleAction=ignore` and `HandleLidSwitch=ignore`.
2. `scripts/host-power-management/user_session_no_suspend_bootstrap.sh` — per-user, no-sudo defensive layer. Idempotent. Safe to source from `start.sh` / `setup.sh` / `bootstrap.sh`.
3. `scripts/host-power-management/check-no-suspend-calls.sh` — static scanner. Exits non-zero on any forbidden invocation.
4. `challenges/scripts/host_no_auto_suspend_challenge.sh` — asserts the running host's state matches layer-1 masking.
5. `challenges/scripts/no_suspend_calls_challenge.sh` — wraps the scanner as a challenge that runs in CI / `run_all_challenges.sh`.

Enforcement: Every project's `CI / run_all_challenges.sh` equivalent MUST run both challenges (host state + source tree). A violation in either channel blocks merge. Adding files to the scanner's `EXCLUDE_PATHS` requires an explicit justification comment identifying the non-host context.

See also: docs/HOST_POWER_MANAGEMENT.md for full background and runbook.

Definition of Done

A change is done when:

1. The code change is committed.
2. All project-level tests pass on a clean clone.
3. All challenges in challenges/scripts/ pass on the running host.
4. Governance docs (CONSTITUTION.md, AGENTS.md, CLAUDE.md) are coherent with the change.

See also

- README.md — project overview, quickstart.
- AGENTS.md — guidance for AI coding agents (Codex, Cursor, etc.).
- CLAUDE.md — guidance specifically for Claude Code.
- docs/HOST_POWER_MANAGEMENT.md — CONST-033 background and runbook.

CONST-035 — Anti-Bluff Tests & Challenges (mandatory; inherits from root)

Tests and Challenges in this submodule **MUST** verify the product, not the LLM's mental model of the product. A test that passes when the feature is broken is worse than a missing test — it gives false confidence and lets defects ship to users. Functional probes at the protocol layer are mandatory:

- TCP-open is the FLOOR, not the ceiling. Postgres → execute SELECT 1. Redis → PING returns PONG. ChromaDB → GET /api/v1/heartbeat returns 200. MCP server → TCP connect + valid JSON-RPC handshake. HTTP gateway → real request, real response, non-empty body.
- Container Up is NOT application healthy. A docker/podman ps Up status only means PID 1 is running; the application may be crash-looping internally.
- No mocks/fakes outside unit tests (already CONST-030; CONST-035 raises the cost of a mock-driven false pass to the same severity as a regression).
- Re-verify after every change. Don't assume a previously-passing test still verifies the same scope after a refactor.
- Verification of CONST-035 itself: deliberately break the feature (e.g. kill <service>, swap a password). The test **MUST** fail. If it

still passes, the test is non-conformant and **MUST** be tightened.

CONST-033 clarification — distinguishing host events from sluggishness

Heavy container builds (BuildKit pulling many GB of layers, parallel podman/docker compose-up across many services) can make the host **appear** unresponsive — high load average, slow SSH, watchers timing out. **This is NOT a CONST-033 violation.** Suspend / hibernate / logout are categorically different events. Distinguish via:

- uptime — recent boot? if so, the host actually rebooted.
- loginctl list-sessions — session(s) still active? if yes, no logout.
- journalctl ... | grep -i 'will suspend\|hibernate' — zero broadcasts since the CONST-033 fix means no suspend ever happened.
- dmesg | grep -i 'killed process\|out of memory' — OOM kills are also NOT host-power events; they're memory-pressure-induced and require their own separate fix (lower per-container memory limits, reduce parallelism).

A sluggish host under build pressure recovers when the build finishes; a suspended host requires explicit unsuspend (and CONST-033 should make that impossible by hardening IdleAction=ignore + HandleSuspendKey=ignore + masked sleep.target, suspend.target, hibernate.target, hybrid-sleep.target).

If you observe what looks like a suspend during heavy builds, the correct first action is **not** "edit CONST-033" but `bash challenges/scripts/host_no_auto_suspend_challenge.sh` to confirm the hardening is intact. If hardening is intact AND no suspend broadcast appears in journal, the perceived event was build-pressure sluggishness, not a power transition.

MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE (User mandate, 2026-04-28)

Forensic anchor — direct user mandate (verbatim):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT**

be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Operative rule: the bar for shipping is **not** "tests pass" but **"users can use the feature."** Every PASS in this codebase MUST carry positive evidence captured during execution that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-based PASS without runtime evidence are all critical defects.

Tests AND Challenges (HelixQA) are bound equally — a Challenge that scores PASS on a non-functional feature is the same class of defect as a unit test that does.

Verification commands

Run before claiming a fix is complete:

```
bash scripts/anti-bluff/bluff-scanner.sh --mode all
bash yole-challenges/scripts/anchor_manifest_challenge.sh
bash yole-challenges/scripts/mutation_ratchet_challenge.sh
```

All three must PASS. Pre-existing bluff hits are tracked in yole-challenges/baselines/bluff-baseline.txt; do not extend the baseline without an explicit justification comment.

Skip-marker convention: // SKIP-OK: #<ticket> (canonical), // ANTI-BLUFF-EXEMPT: <reason> (synonym).

Submodule Decoupling & Reusability — Mandatory

Status: Mandatory. Non-negotiable.

Rule: This repository is a **shared submodule** consumed by multiple independent consumer projects. Its value depends on staying **fully decoupled and reusable**. No change in this repository may introduce coupling that breaks its standalone reusability for any consumer.

Prohibited inside this repository:

1. Hardcoding any specific consumer project's name, paths, platform list, version strings, release-naming conventions, branding, or feature names.
2. import / dependency on any consumer-project namespace, package, or build coordinate.
3. Embedding consumer-project-specific governance, rule numbering, or release cadence into this repository's CONSTITUTION.md / CLAUDE.md / AGENTS.md.

4. Assuming this repository is consumed by a particular CLI, build system, language toolchain version, or target architecture beyond what its public interface documents.

Required inside this repository:

1. All public surfaces (APIs, CLIs, configuration files, environment variables, scripts) **MUST** be expressed in terms of THIS repository's own domain — not any consumer's.
2. Governance **MUST** describe responsibilities and contract from THIS repository's perspective. Consumer projects appear as illustrative examples at most, never as load-bearing requirements.
3. Cross-project rules adopted from a consumer (such as a cross-platform impact mandate) **MUST** be phrased generically — "every consuming project's full platform matrix" — and never hardcode any single consumer's matrix.

Why: Repositories like this one have shipped changes in the past where one consumer's platform list, feature names, or rule numbering leaked into shared-repo governance — and then collided at merge time with another consumer's parallel work, leaving the repository unmergeable until manual conflict resolution stripped the consumer-specific text back out. Decoupling is the only mechanism that preserves this repository's value as shared infrastructure.

Recursive scope: any submodule this repository consumes inherits the same decoupling+reusability rule. Third-party upstream submodules that this repository merely vendors (e.g. open-source tools under a tools/opensource/ tree, if present) are explicitly out of scope — we are not their owners.

Article XI §11.9 — Anti-Bluff Forensic Anchor (cascaded from parent CONSTITUTION.md)

Verbatim user mandate (2026-04-29, reasserted multiple times across 2026-05): *"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"*

Operative rule: **The bar for shipping is not "tests pass" but "users can use the feature."** Every PASS in this codebase **MUST** carry positive runtime evidence captured during execution. Metadata-only / configuration-only / absence-of-error / grep-based

PASS without runtime evidence are critical defects regardless of how green the summary line looks. No false-success results are tolerable.

This anchor MUST remain in this submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md alongside CONST-047 — see the parent repository's CONSTITUTION.md for the full text.

CONST-048: Full-Automation-Coverage Mandate (cascaded from constitution submodule §11.4.25)

Verbatim user mandate (2026-05-15): "Make sure that every feature, every functionality, every flow, every use case, every edge case, every service or application, on every platform we support is covered with full automation tests which will confirm anti-bluff policy and provide the proof of fully working capabilities, working implementation as expected, no issues, no bugs, fully documented, tests covered! Nothing less than this does not give us a chance to deliver stable product! This is mandatory constraint which MUST BE respected without ignoring, skipping, slacking or forgetting it!"

No feature / functionality / flow / use case / edge case / service / application on any supported platform of HelixCode may be considered deliverable until covered by automation tests proving six invariants: (1) anti-bluff posture (CONST-035) with captured runtime evidence; (2) proof of working capability end-to-end on target topology (no mocks beyond unit tests — see CONST-050); (3) implementation matches documented promise; (4) no open issues/bugs surfaced — cross-checked against §11.4.15 / §11.4.16 trackers; (5) full documentation in sync per §11.4.12; (6) four-layer test floor per §1 (pre-build + post-build + runtime + paired mutation).

Consuming projects MUST publish a coverage ledger (feature × platform × invariant-1..6 × status) regenerated as part of the release-gate sweep. Gaps tracked per §11.4.15 (UNCONFIRMED: / PENDING_FORENSICS: / OPERATOR-BLOCKED: with §11.4.21 audit) — rows that quietly omit a platform are CONST-048 violations.

Cascade requirement: This anchor (verbatim or by CONST-048 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. No escape hatch. See constitution submodule Constitution.md §11.4.25 for the full mandate.

CONST-049: Constitution-Submodule Update Workflow Mandate (cascaded from constitution submodule §11.4.26)

Verbatim user mandate (2026-05-15): *"Every time we add something into our root (constitution Submodule) Constitution, CLAUDE.MD and AGENTS.MD we MUST FIRST fetch and pull all new changes / work from constitution Submodule first! All changes we apply MUST BE committed and pushed to all constitution Submodule upstreams! In case of conflict, IT MUST BE carefully resolved! Nothing can be broken, made faulty, corrupted or unusable! After merging full validation and verification MUST BE done!"*

Before ANY modification to constitution/Constitution.md, constitution/CLAUDE.md, or constitution/AGENTS.md, the agent or operator MUST execute the following 7-step pipeline in order:

1. **Fetch + pull first** inside the constitution submodule worktree — every configured remote fetched, then `git pull --ff-only` (or `--rebase` if non-FF; NEVER `--strategy=ours` / `--allow-unrelated-histories` without explicit authorization).
2. **Apply the change** with §11.4.17 classification + verbatim mandate quote.
3. **Validate before commit** — `meta_test_inheritance.sh` (or equivalent), no merge-conflict markers, cross-file consistency.
4. **Commit + push to ALL upstreams** — governance files only (NEVER `git add -A`); push to every configured remote. One-upstream commit = CONST-049 violation (also CONST-038/§6.W and §2.1).
5. **Conflict resolution** preserving union of governance content. Force-push to bypass conflicts is FORBIDDEN (CONST-043 / §9.2).
6. **Post-merge validation** — `git submodule update --remote --init` + re-run cascade verifier (CONST-047) confirming the new clause reaches every owned submodule.
7. **Bump consuming project pointer** — `.gitmodules-tracked` submodule pointer advanced to the new constitution HEAD in the SAME commit as cascade work.

Cascade requirement: This anchor (verbatim or by CONST-049 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a force-push without CONST-043 / §9.2 authorization. No escape hatch. See constitution submodule Constitution.md §11.4.26 for the full mandate.

CONST-050: No-Fakes-Beyond-Unit-Tests + 100%-Test-Type-Coverage Mandate (cascaded from constitution submodule §11.4.27)

Verbatim user mandate (2026-05-15): *"Mocks, stubs, placeholders, TODOs or FIXMEs are allowed to exist ONLY in Unit tests! All other test types MUST interact with real fully implemented System! No fakes, empty implementations or bluffing is allowed of any kind! All codebase of the project MUST BE 100% covered with every supported test type: unit tests, integration tests, e2e tests, full automation tests, security tests, ddos tests, scaling tests, chaos tests, stress tests, performance tests, benchmarking tests, ui tests, ux tests, Challenges (fully incorporating our Challenges Submodule — <https://github.com/vasic-digital/Challenges>). EVERYTHING MUST BE tested using HelixQA (fully incorporating HelixQA Submodule — <https://github.com/HelixDevelopment/HelixQA>). HelixQA MUST BE used with all possible written tests suites (test banks) for every applications, service, platform, etc and execution of the full HelixQA QA autonomous sessions! All required dependency Submodules MUST BE added into the project as well (fully recursive!!!)."*

Two cooperating invariants:

(A) No-fakes-beyond-unit-tests. Mocks, stubs, fakes, placeholders, TODO, FIXME, "for now", "in production this would", or empty-implementation patterns are PERMITTED only in unit-test sources (*_test.go files invoked without the integration build tag; HelixCode/tests/unit/; etc.). Every other test type — integration, E2E, full automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA — MUST exercise the real, fully implemented HelixCode system against real infrastructure (real PostgreSQL, real Redis, real LLM endpoints, real containers, real captured devices). Production code (anything under HelixCode/cmd/, HelixCode/applications/, HelixCode/internal/<pkg>/<file>.go not ending *_test.go) MUST NOT import from HelixCode/internal/mocks/.

(B) 100% test-type coverage. HelixCode's codebase MUST be covered by every supported test type the domain warrants:

- **Unit** — fast, isolated, mocks permitted per (A).
- **Integration** — multi-component, no mocks, real backing services.
- **End-to-end (E2E)** — full user-flow exercise on target topology.

- **Full automation** — orchestrated suites exercising every feature × platform combination (CONST-048 coverage ledger).
- **Security** — authn/authz boundaries, CONST-042 secret-leak scans, input-fuzzing, dependency-CVE scanning, threat-model verification.
- **DDoS** — request-flood resilience at advertised throughput tier.
- **Scaling** — horizontal + vertical scale behaviour under linear load growth.
- **Chaos** — controlled failure injection (network partition, process kill, disk full, clock skew).
- **Stress** — sustained load above advertised tier.
- **Performance** — latency / throughput / tail-latency invariants vs SLO baselines.
- **Benchmarking** — micro + macro suites with historical p95-drift detection.
- **UI** — visual-regression + DOM-state + interaction-flow coverage on every target platform's UI surface.
- **UX** — flow-correctness + accessibility + i18n + visual-cue ordering (§11.4.23 composition).
- **Challenges** — vasic-digital/Challenges submodule (at ./Challenges/) fully incorporated; per-feature Challenge scripts with captured runtime evidence.
- **HelixQA** — HelixDevelopment/HelixQA submodule (at ./HelixQA/) fully incorporated; ALL written test banks executed; full autonomous QA sessions run as part of release gates with captured wire evidence per check.

Required dependency submodules (recursive per CONST-047):

- Challenges — git@github.com:vasic-digital/Challenges.git — incorporated at ./Challenges/.
- HelixQA — git@github.com:HelixDevelopment/HelixQA.git — incorporated at ./HelixQA/.
- Any additional functionality submodules under vasic-digital/* / HelixDevelopment/* orgs that HelixCode depends on — incorporate rather than duplicate work the orgs already maintain.

Submodule pointers **MUST** be bumped to upstream HEAD in the SAME commit as any dependent cascade work (CONST-049 step 7). Pointer drift = CONST-050 violation.

Cascade requirement: This anchor (verbatim or by CONST-050 ID reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. No escape hatch. See constitution submodule Constitution.md §11.4.27 for the full mandate.

CONST-051: Submodules-As-Equal-Codebase + Decoupling + Dependency-Layout Mandate (cascaded from constitution submodule §11.4.28)

Verbatim user mandate (2026-05-15): *"All existing Submodules in the project that we are controlling and belong to some our organizations (vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory - we can ALWAYS check dynamically using GitHub and GitLab CLIs) are equal parts of the project's codebase! We MUST work on that code as much as we do with main project's codebase! All on equal basis! Equally important! We MUST take it into the account, analyze it, extend it, create missing tests, do full testing of it, fill the gaps (if any), fix any issues that we discover or they pop-up, write and extend the documentation, user guides, manulas, diagrams, graphs, SQL definitions, Website(s) and all other relevant materials! We MUST NEVER modify Submodules to bring into them any project specific context since they all MUST BE ALWAYS fully decoupled, project not-aware, fully reusable and modular (by any other project(s)), completely testable! All Submodule dependencies that are used by Submodule MUST BE accessed from the root of the project! We MUST NOT have nested Submodule dependencies but accessing each from proper location from the root of the project - directly from project's root project_name/submodule_name or some more proper structure project_name/submodules/submodule_name!"*

Three cooperating invariants apply to every HelixCode-owned submodule (those whose upstream origin lives under vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory, or any subsequently authorised org):

(A) Equal-codebase. Every owned-by-us submodule is an **equal part** of HelixCode's codebase. The same engineering practice — analysis, extension, test creation, gap-filling, bug-fix, documentation (user manuals, guides, diagrams, graphs, SQL definitions, website pages, all materials) — applies to each owned submodule on equal basis. A round of work that improves only HelixCode's main while leaving an owned-submodule deficiency unaddressed is a CONST-051 violation, severity-equivalent to a §11.4 PASS-bluff at the project-scope layer. The §11.4.25 / CONST-048 coverage ledger MUST list every owned submodule as an in-scope target.

(B) Decoupling / reusability. Owned submodules MUST remain fully decoupled from HelixCode (and any other consuming project). No HelixCode-specific context, hardcoded paths, hostnames,

asset names, or runtime assumptions may be introduced into an owned submodule's source tree. When a submodule needs information from HelixCode, the honest path is configuration injection (env var, config file, constructor parameter) — never a hardcoded reach into the parent's tree. Every owned submodule **MUST** be project-not-aware, fully reusable, modular, and completely testable as a standalone repository.

(C) Dependency-layout. Every dependency that an owned submodule consumes **MUST** be accessible from HelixCode's root at one of two canonical paths:

- `<repo_root>/<submodule_name>/` (flat layout — current HelixCode layout for Challenges, HelixQA, Containers, Security, etc.)
- `<repo_root>/submodules/<submodule_name>/` (grouped layout — alternate)

Nested own-org submodule chains are FORBIDDEN. A submodule **MUST NOT** have its own `.gitmodules` entries pulling in further owned-by-us repos. Every dependency required by submodule X is added to HelixCode's root at the canonical path; X reaches it via documented import / SDK path / runtime resolver — never via its own nested submodule pointer. Third-party submodules (not under our orgs) are exempt — they **MAY** appear at any depth.

The owned-org list is dynamically discoverable at any time via `gh org list` / `glab CLIs` or the orgs' public APIs.

Cascade requirement: This anchor (verbatim or by CONST-051 ID reference) **MUST** appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to a §11.4 PASS-bluff at the codebase-completeness layer. No escape hatch. See constitution submodule `Constitution.md` §11.4.28 for the full mandate (audit gates, mutation pairs, workflow integration).

Amendment Process

Constitution amendments require:

1. Written proposal with rationale
 2. Challenge demonstrating the need
 3. 72-hour review period
 4. Approval by project architect
 5. Update to all submodule governance files
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This Constitution is the supreme law of the HelixCode project. No code, test, or process may contradict it.

CONST-052: Lowercase-Snake_Case-Naming Mandate (cascaded from constitution submodule §11.4.29)

Verbatim user mandate (2026-05-15): *"naming convention for Submodules and directories (applied deep into hierarchy recursively) - all directories and Submodules MSUT HAVE lowercase names with space separator between the words of '_' character (snake-case)! All existing Submodules and directories which are not following this rule MUST BE renamed! However, since this will most likely break some of the functionalities renaming we do MUST BE applied to all references to particular Submodule or directory! ... There MUST BE reasonable exceptions for this rules - source code for programming languages or Submodules which apply different naming convention - Android, Java, Kotlin and others. ... Upstreams directory which all of our projects and Submodules have MUST BE renamed to the lowercase letters too, however root project containing the install_upstreams system command (it is exported in out paths in our .bashrc or .zshrc) MUST BE updated to fully work with both Upstreams and upstreams directory. ... NOTE: Rules lowercase / snake-case do apply to all project files as well and references to it and from them!"*

Every directory, submodule, and file in HelixCode MUST use lowercase snake_case names. Existing non-compliant names (HelixCode/, Challenges/, Containers/, HelixAgent/, HelixQA/, Security/, Github-Pages-Website/, Upstreams/, Dependencies/, etc.) MUST be renamed as part of the phased migration opened by this clause. Every reference (configs, docs, links, source-code imports, governance files) MUST be updated atomically with the rename — reference drift after a rename is a CONST-052 violation of equal severity to the rename itself.

Common-sense exceptions (technology-preserving): language-mandated case for Java/Kotlin/Android/Apple/C#/Swift INSIDE the language root (submodule root follows our convention; subtree follows language convention); vendor/upstream third-party submodules keep upstream names; build artefacts (node_modules, __pycache__, .git, target, build, bin) keep tool-mandated names. The test "does renaming break the technology?" trumps the rule.

Upstreams/ → upstreams/ transition: the constitution submodule's install_upstreams.sh (exported via .bashrc/.zshrc) supports BOTH Upstreams/ and upstreams/ directory layouts (commit 45d3678 of the constitution submodule); lowercase wins when both present.

Test coverage of renames (per CONST-050(B)): every rename batch ships with (i) regression test verifying every reference now resolves, (ii) full test-type matrix run post-rename, (iii) anti-bluff wire-evidence captured.

Phased execution per the operator's explicit instruction: comprehensive brainstorming → phase-divided plan → fine-grained tasks/subtasks → every change covered by every applicable test type. §11.4.20 subagent delegation for cross-cutting rename sweeps.

Cascade requirement: This anchor (verbatim or by CONST-052 ID reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the reference-integrity layer. No escape hatch beyond the common-sense exceptions enumerated above. See constitution submodule Constitution.md §11.4.29 for the full mandate.

CONST-053: .gitignore + No-Versioned-Build-Artifacts Mandate (cascaded from constitution submodule §11.4.30)

Verbatim user mandate (2026-05-15): *"every project module, every Submodule, every service and application MUST HAVE proper .gitignore file! We MUST NOT git version build artifacts, cache files, tmp files, main .env file(s) or any files containing sensitive data, API keys or token! Any build derivative which we can recreate by executing proper mechanism for generating MUST NOT be versioned! We MUST pay attention what is going to be committed every time we are preparing to execute commit! If any violation is detected it MUST be fixed before commit is executed!"*

Every project module, owned-by-us submodule, service, and application **MUST** ship a proper .gitignore. Forbidden-from-version-control classes:

1. **Build artefacts:** /bin/, /build/, /dist/, /out/, target/, *.exe, *.dll, *.so, *.dylib, *.a, *.o, *.class, *.pyc, generator-produced files when the generator is committed.
2. **Cache files:** __pycache__/, .pytest_cache/, .mypy_cache/, .ruff_cache/, node_modules/, .next/, .cache/, .gradle/, .terraform/, language-server caches.
3. **Temp files:** *.tmp, *.swp, *~, .DS_Store, Thumbs.db, *.orig, *.rej.
4. **Sensitive-data files:** .env, .env.* (allow .env.example placeholder only — no real secrets even as examples), *.pem, *.key, *.crt, id_rsa*, id_ed25519*, .netrc, secrets/, api_keys.sh.

5. **Generated reports/logs:** *.log, coverage.out, htmlcov/, runtime captures unless reference assets.
6. **OS/IDE personal state:** .idea/, .history/, .vscode/ (except shared settings).

Anti-bluff invariant: .gitignore line alone is not sufficient — no file matching the forbidden patterns may be CURRENTLY TRACKED. A tracked *.log despite the ignore-line is a violation of equal severity to no ignore-line at all.

Pre-commit attention: every commit author (human OR agent) MUST inspect `git diff --staged + git status` BEFORE executing the commit. Forbidden-class hits abort the commit until fixed (unstage, add to .gitignore, scrub if already-tracked). Gate CM-GITIGNORE-PRECOMMIT-AUDIT + paired mutation.

Secret-leak intersection (CONST-042 / §11.4.10): a .env leak is BOTH a CONST-053 and a CONST-042 violation; rotation + post-mortem required.

Recreatable-content test: if a documented mechanism regenerates the file from sources, it is a build derivative and MUST be ignored. The committed sources MUST include the generator.

Cascade requirement: This anchor (verbatim or by CONST-053 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the repository-hygiene layer. See constitution submodule Constitution.md §11.4.30 for the full mandate.

CONST-054: Submodule-Dependency-Manifest Mandate (cascaded from constitution submodule §11.4.31)

Verbatim user mandate (2026-05-15): "*We MUST HAVE mechanism for each Submodule to determine / know what are its Submodule dependencies so new projects or palces we are incorporate them can add these Submodules to the project root and make them available! Suggested idea is configuration file with expected Submodules Git ssh urls perhaps? New project can read it, and recursively add each Submodule to the root of the project and install / expose it to veryone.*"

Every owned-by-us submodule MUST ship helix-deps.yaml at its root declaring its own-org dependencies. Schema: `schema_version`, `deps: [{name, ssh_url, ref, why, layout: flat|grouped}],` `transitive_handling.{recursive,conflict_resolution},` `language_specific_subtree`. Tooling: `incorporate-submodule <ssh-url>` adds the submodule at the parent project's canonical path

(CONST-051(C)), reads `helix-deps.yaml`, recurses for each declared dep, aborts on conflicting refs, emits `<root>/helix-manifest.yaml` audit record.

Anti-bluff guarantee: every manifest paired with a Challenge that bootstraps a throwaway consuming project, runs incorporate-submodule, asserts produced layout matches the manifest, runs the submodule's own tests against the bootstrapped layout, captures wire evidence per §11.4.2. A manifest without this proof is a CONST-054 violation.

§11.4.31 / CONST-054 is the **operational complement** of CONST-051(C): nested own-org submodule chains are FORBIDDEN — manifests are the bridge that lets consumers reconstruct the dependency graph at the parent root.

Cascade requirement: This anchor (verbatim or by CONST-054 ID reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to §11.4 PASS-bluff at the dependency-graph layer. See constitution submodule Constitution.md §11.4.31 for the full mandate.

CONST-055: Post-Constitution-Pull Validation Mandate (cascaded from constitution submodule §11.4.32)

Verbatim user mandate (2026-05-15): *"Every time we fetch and pull new changes on constitution Submodule we MUST process the whole project and all Submodule (deep recursively) for validation and verification taht every single rule or mandatory constraint is followed and respected! If it is not, IT MUST BE!"*

Whenever a project's constitution submodule is fetched + pulled with any content change, the project **MUST** run `scripts/verify-all-constitution-rules.sh` BEFORE the new constitution HEAD is treated as canonical for any other work. The sweep re-runs the governance-cascade verifier AND every implementable rule gate (CONST-053 `.gitignore` audit, CONST-051(C) nested-own-org-chain audit, CONST-052 case audit, CONST-050(A) mock-from-production audit, CONST-035 anti-bluff smoke, etc.) against the post-pull tree. Failures populate the project's Issues tracker per §11.4.15 (Status: Reopened, Type: Bug); closure requires positive-evidence per §11.4.

Pull-time invocation: `git submodule update --remote constitution` triggers the sweep automatically (post-update hook OR commit-wrapper invocation). Operator-explicit manual invocation also available.

Anti-bluff: the sweep's own meta-test (paired mutation per §1.1) plants a known violation of each enforced gate and asserts the sweep reports FAIL for the planted gate. A sweep that exits PASS without running every implementable gate is a CONST-055 violation.

CONST-055 is the **enforcement engine** for every other §11.4.x and CONST-NNN rule — without it, new rules cascade as anchors but never get enforced.

Cascade requirement: This anchor (verbatim or by CONST-055 ID reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to §11.4 PASS-bluff at the constitutional-enforcement layer. See constitution submodule Constitution.md §11.4.32 for the full mandate.

CONST-056: Mandatory install_upstreams on clone/add Mandate (cascaded from constitution submodule §11.4.36)

Verbatim user mandate (2026-05-15): "Every Submodule or Git repository we add or clone MUST BE upstreams installed using Upstreamable utility which MUST BE available through exported paths of the host system (in .bashrc or .zshrc) using install_upstreams command executed from the root of the cloned (added) repository - only if in it is Upstreams or upstreams directory present with bash script files (recipes) for all repository's upstreams!"

Every clone / add of a Git repository under HelixCode **MUST** be followed by install_upstreams invocation from the repository's root IF its tree contains upstreams/ (or legacy Upstreams/ per CONST-052 transition) populated with *.sh recipe files. The utility (installed on operator's PATH via .bashrc/.zshrc; implementation in the constitution submodule's install_upstreams.sh — already supports BOTH directory names since constitution commit 45d3678) reads the recipe files, configures every declared upstream as a named git remote, and fans out origin push URLs.

Skipping the invocation when upstreams/ is present silently breaks §2.1 (multi-upstream push is the norm) — the next push lands on only one upstream. Gate CM-INSTALL-UPSTREAMS-ON-CLONE + paired mutation. Automation: the future incorporate-submodule per CONST-054 auto-invokes; manual invocation supported. Pre-commit check: `git remote -v | grep -c push` reports expected count.

Cascade requirement: This anchor (verbatim or by CONST-056 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.36 for the full mandate.

CONST-057: Type-aware Closure-Status Vocabulary (cascaded from constitution submodule §11.4.33)

Every project tracking work items by Type per §11.4.16 MUST close them with the Type-appropriate terminal **Status** value, drawn from this 3-element closed map:

Item Type	Closure Status value
Bug	Fixed (→ Fixed.md)
Feature	Implemented (→ Fixed.md)
Task	Completed (→ Fixed.md)

The (→ Fixed.md) suffix is preserved across all three so the existing migration-discipline tooling (atomic Issues.md → Fixed.md move per §11.4.19) keeps working without per-Type branching. Generators (generate_issues_summary.sh, generate_fixed_summary.sh, the §11.4.23 colorizer) MUST treat the three terminal values as semantically equivalent (all "closed, positive evidence captured") while preserving the literal in the emitted document.

Closing a Feature with Fixed (→ Fixed.md) or a Task with Implemented (→ Fixed.md) is a CONST-057 violation. Gate CM-CLOSURE-VOCAB-TYPE-AWARE walks every Fixed.md heading + every Issues.md heading whose **Status** is one of the three terminal values and asserts the Status-Type match. Composes with §11.4.15 / §11.4.16 / §11.4.19 / §11.4.23.

Cascade requirement: This anchor (verbatim or by CONST-057 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.33 for the full mandate.

CONST-058: Reopened-Source Attribution Mandate (cascaded from constitution submodule §11.4.34)

Every Issues.md (or equivalent project tracker) heading whose ****Status:**** is Reopened MUST carry, within 8 non-blank lines of the heading, a ****Reopened-Details:**** line capturing four sub-facts:

- **By:** AI or User (source-of-truth observer who flipped the status). AI covers in-loop reopens (test failure, gate regression, captured-evidence retrospect). User covers operator-side observations (manual testing, end-user report, design reconsideration).
- **On:** ISO date (YYYY-MM-DD).
- **Reason:** one-line cause classification — chosen from the closed vocabulary { test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered }. Other values permitted with explicit Reason: <free text> annotation but the closed list MUST be tried first.
- **Evidence:** path to or short description of the captured artefact justifying the reopen — log file, recording, gate failure ID, operator quote, etc. Reopens without evidence are §11.4.6 / §11.4.7 violations (demotion from Fixed requires captured evidence under the conditions that re-exposed the defect).

The Issues_Summary.md Status column MUST distinguish the four Reopened sub-states by source so a sweep query for "reopens by AI in the last 30 days" is mechanically possible. Suggested column rendering: Reopened (AI: test-failed) vs Reopened (User: manual-testing). Gate CM-ITEM-REOPENED-DETAILS mirrors CM-ITEM-OPERATOR-BLOCKED-DETAILS (§11.4.21 walk pattern). Composes with §11.4.6 / §11.4.7 / §11.4.15 / §11.4.21.

Cascade requirement: This anchor (verbatim or by CONST-058 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.34 for the full mandate.

CONST-059: Canonical-Root Inheritance Clarity (cascaded from constitution submodule §11.4.35)

The **constitution submodule's** three files (constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md) ARE the **canonical root** (also called the **parent** files). They contain only universal rules per §11.4.17.

The consuming project's **repository-root files** (<project-root>/CLAUDE.md, <project-root>/AGENTS.md, optionally <project-root>/Constitution.md) are **consumer extensions**. They MUST start with the inheritance pointer (either the Claude-Code native @constitution/CLAUDE.md import or the portable ## INHERITED FROM constitution/CLAUDE.md heading). They contain only project-specific rules per §11.4.17.

When in doubt about which file to edit: universal rule → constitution submodule's file; project-specific rule → consumer's file. Default consumer-side when uncertain (§11.4.17 — narrower scope is cheap to widen).

Terminology: "the parent CLAUDE.md" / "the root Constitution" → constitution-submodule file at constitution/<filename>; "the project CLAUDE.md" / "this project's AGENTS.md" → consumer-side file at <project-root>/<filename>.

No silent demotion or silent promotion. Moving a rule between layers MUST be a visible commit — git mv of a section if it's a clean clone, or explicit
Lifted from <project> to constitution per §11.4.35 / Demoted from constitution to <project> per §11.4.35 commit-message annotation.

Gate CM-CANONICAL-ROOT-CLARITY verifies (a) consumer's CLAUDE.md opens with the inheritance pointer, (b) constitution submodule's three files are present at the expected path, (c) no ## INHERITED FROM block in the constitution submodule's own files (those ARE the source-of-truth, not consumers). Composes with §11.4.17.

Cascade requirement: This anchor (verbatim or by CONST-059 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.35 for the full mandate.

CONST-060: Fetch-before-edit Mandate (cascaded from constitution submodule §11.4.37)

Verbatim user mandate (2026-05-15): *"Make sure that feedback_fetch_before_edit memory rule is part of our constitution Submodule - the root Constitution, AGENTS.MD and CLAUDE.MD. Validate and verify that Project-Toolkit and all Submodules do inherit all of them! Follow the constitution Submodule documentation for details."*

The FIRST git-touching action of every session, on every consuming project (owned or third-party), MUST be:

```
git fetch --all --prune
git log --oneline HEAD..@{u}
git submodule foreach --recursive 'git fetch --all --prune --
quiet'
```

If HEAD..@{u} is non-empty, integrate the upstream changes BEFORE any local edit. Acting on stale local state produces three failure modes documented in the originating §11.4.37 incident (multi-agent / parallel-session work): (1) **redundant work** — the agent re-does what a parallel session already finished, (2) **false confidence** — completion reports for already-done work, (3) **divergent history** — duplicate sibling commits that double the conflict surface on next push.

Anti-bluff invariant: the fetch+log check MUST produce captured evidence — the actual HEAD..@{u} output, even if empty. Skipping the check on the basis of "I just fetched" or "nothing could have changed in the last N minutes" is a §11.4.6 (no-guessing) violation: the remote state is not knowable without a fetch.

Cascade requirement: This anchor (verbatim or by CONST-060 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to §11.4 PASS-bluff at the parallel-session-coordination layer. See constitution submodule Constitution.md §11.4.37 for the full mandate.

Anti-Bluff End-User Quality Guarantee (Escalated via HelixConstitution)

Canonical authority: HelixConstitution/Constitution.md §7.1 and §11.4. This section is this submodule's binding pointer to those universal clauses.

**Forensic anchor — verbatim operator mandate
(2026-04-28):**

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product! This MUST BE part of Constitution of our project, its CLAUDE.MD and AGENTS.MD if it is not there already, and to be applied to all Submodules's Constitution, CLAUDE.MD and AGENTS.MD as well (if not there already)!"

Operative rule: the bar for shipping is **not** "tests pass" but **"users of any consuming project can use the feature."** Every PASS MUST carry positive runtime evidence captured during exe-

cution that the feature actually works end-to-end. Metadata-only PASS, configuration-only PASS, "absence-of-error" PASS, and source-grep-only PASS without runtime evidence are critical defects.

Required minimum evidence per test category:

Category	Minimum evidence
Unit tests	Real inputs + mutation-verified assertions (stub-swap must cause FAIL)
Integration tests	Real subsystems only; mocks only at honest external boundaries
E2E / Challenge tests	Per-test PASS/FAIL lines + log-file artefact path; RUNTIME layer mandatory

Decoupling constraint: this rule applies to every consuming project's full platform matrix — specific platform names are not hardcoded here.

CONST-061: Pre-Force-Push Merge-First Mandate (cascaded from constitution submodule §11.4.41)

Verbatim user mandate (2026-05-17): *"make sure we bring everything from branches to our side before forc push is done! Afer everything is safely and fully merged and all potential conflicts (if any) resolved, then do force push! make sure nothing isnlost, broken or corrupted on bith sides! add these rules in our root Constitution, CLAUDE.MD, AGENTS.MD (constitution Submodule) if itnis not added already! Extremely important rules and mandatory constraints we MUST HAVE and fully respect!"*

Any force-push (--force, --force-with-lease, +<ref>, equivalent history-rewrite) authorised under CONST-043 MUST be preceded by a mechanical 4-step merge-first pipeline:

1. **Fetch every remote** — git fetch --all --prune --tags against origin + every upstream; capture output.
2. **Integrate every divergent commit locally** — rebase / merge / operator-confirmed cherry-pick per appropriate strategy for every non-empty HEAD..<<remote>/<branch> range.
3. **Audit the integrated tree** — no conflict markers anywhere (grep -rn '^<<<<<<< \|^===== \$\|^>>>>>>>' returns empty in governance + source + test files); no file silently dropped; previously-passing tests still pass; captured-evidence artefacts still validate.

4. **Force-push** — only after steps 1-3 produce clean integration evidence: `git push --force-with-lease` (NEVER `--force` alone unless authorised per §9.2 sub-clause 6).

Two-gate composition with CONST-043. §11.4.41 does NOT relax CONST-043's operator-approval requirement — it adds a SECOND mechanical gate. CONST-043 alone authorises a push that loses remote work; §11.4.41 alone risks pushing without operator awareness. Both required.

Three failure modes prevented: (a) remote-side content loss when parallel sessions land work between fetches; (b) stale-state acts when `--force-with-lease` reads stale local refs without prior fetch; (c) conflict-driven corruption when markers get committed verbatim (observed 2026-05-17 in `helix_qa` + containers governance files).

Verification artefact: every governed force-push emits a `docs/changelogs/<tag>.md` "Force-push merge-first audit" section capturing fetch output, per-remote divergence log, integration strategy, conflict-marker scan, test delta, push output with lease SHA, + CONST-043 authorisation quote. Gate CM-FORCE-PUSH-MERGE-FIRST + paired mutation.

Cascade requirement: This anchor (verbatim or by CONST-061 ID reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to a §11.4 PASS-bluff at the remote-data-integrity layer. See constitution submodule `Constitution.md` §11.4.41 for the full mandate.

CONST-068: Shell-script target-shell-parseability mandate (cascaded from constitution submodule §11.4.67)

Verbatim user mandate (2026-05-19): *"any issue we spot must be fixed, bash scripts as well if they are broken!" + "Make sure that this is mandatory rule!"*

Verbatim 2026-05-19 operator mandate: *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"*

Every committed shell script **MUST** be parseable by its target interpreter (sh -n for /bin/sh, bash -n for /bin/bash, etc.) **AND** **MUST** declare a shebang matching its actual syntax usage. Bash-only constructs (>(...), <(...), [[]], <<<, arrays, \${var^^}, etc.) used in scripts that may be invoked via sh script.sh **MUST** be wrapped in eval so the parser sees only a string (target shells like mksh parse the entire script before executing — runtime guards cannot save a parse-time rejection). Honest shebangs only: #!/bin/bash only if bash actually expected; #!/bin/sh requires POSIX-clean body. Fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51. Pre-build gate CM-SCRIPT-TARGET-SHELL-PARSEABLE runs sh -n on every in-scope script. No escape hatch — no --skip-parseability-check, --bash-only-script, --runtime-guard-suffices flag.

Cascade requirement: This anchor (verbatim or by CONST-068 ID reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.67 for the full mandate.

§11.4.68 — Positive Sink-Side / Downstream Evidence Mandate (cascaded from constitution submodule §11.4.68)

Verbatim user mandate (2026-05-20): *"We still do not hear any audio played from D3 device! Arvus Web Dashboard when we play music from D3 shows nothing for Codec In Use! This **MUST BE** investigated and fixed! How come we passed the tests with Arvus validation? What were values for the Codec In Use field? Empty means nothing! This is not working! It **MUST BE FIXED, TESTED AND VERIFIED WITH FULL AUTOMATION TESTING ASAP!!!**"*

A test that asserts audio or video routing **PASS** **MUST** capture and verify **positive sink-side or downstream evidence** — never config-only, never metadata-only, never PCM-open-state-only. At least one of the closed enumeration **MUST** be captured for every audio/video routing **PASS**: (1) sink-side codec-state with non-empty Codec-In-Use matching the expected codec regex; (2) strictly-positive PCM frames-written delta from /proc/asound/.../status hw_ptr; (3) ALSA ELD/EDID-Like-Data showing negotiated channel count + format; (4) ffprobe-on-captured-mp4 with non-zero frame count + expected codec/resolution/fps; (5) recording-analyzer event match per §11.4.2/§11.4.5; (6) tinycap RMS amplitude above the line-level floor. Empty / <unreachable> / <N.E.> / <None> placeholders are **NOT** positive evidence; a missing-but-required sink is

OPERATOR-BLOCKED (release-blocker), never SKIP, never PASS. No escape hatch — no --skip-sink-evidence, --allow-empty-codec, --sink-unreachable-is-pass, --metadata-only-suffices flag exists.

Cascade requirement: This anchor (verbatim or by §11.4.68 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the sink-side-evidence layer. **Canonical authority:** constitution submodule Constitution.md §11.4.68 for the full mandate.

§11.4.70 — Subagent-Driven Execution Is The Default (cascaded from constitution submodule §11.4.70)

Verbatim user mandate (2026-05-20): *"Always do if possible Subagent-driven! Add this into our root (constitution Submodule) Constitution.md, CLAUDE.md and AGENTS.md. This should be the default choice ALWAYS!"*

When executing implementation plans (or any task-decomposed execution flow), the **default execution model is subagent-driven** per superpowers:subagent-driven-development. Inline execution is permitted ONLY when (a) the task is trivial AND fits a single sub-300-line edit, OR (b) the operator explicitly requests inline at brainstorm-handoff time. Subagent-driven is the default because it gives isolated context per task, naturally enforces two-stage review, is parallel-PWU compatible (§11.4.58), creates an anti-bluff seam (§11.4), and survives operator absence. No escape hatch — --inline-execution-required, --no-subagents, --monolithic-execution are NOT permitted flags. Skipping subagent-driven for non-trivial work without recorded operator authorisation is itself a §11.4 PASS-bluff.

Cascade requirement: This anchor (verbatim or by §11.4.70 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the execution-model layer. **Canonical authority:** constitution submodule Constitution.md §11.4.70 for the full mandate.

§11.4.71 — Pre-Push Fetch + Investigate + Integrate Mandate (cascaded from constitution submodule §11.4.71)

Verbatim user mandate (2026-05-20): *"before pushing changes to any upstream for any repository - main repo or Submodule, we MUST fetch and pull all changes. Once these are obtained WE MUST investigate what is different compared to head position we were on last time before fetching and pulling new changes! We MUST understand what is done and for what purpose, easpecially how that does affect our project and our System in general! Any mandatory changes or improvements required by fresh changes we just have brought in MUST BE incorporated, covered with all supported types of the tests which will produce as a result of its success execution REAL PROOFS of working for all components and functionalities covered and work fully in anti-bluff manner!"*

The everyday-push variant of §11.4.41. EVERY push (every repository — main + every submodule) MUST follow the 5-step cycle: (1) fetch all remotes (`git fetch --all --prune --tags`, capture stdout); (2) pull all upstream branches whose tip differs, resolving conflicts per consumer judgment (never auto---ours/--theirs); (3) investigate the diff vs OUR previous HEAD — read EVERY foreign commit's body, understand what/why/how-it-affects-our-system; (4) integrate mandatory changes with §11.4.4(b) four-layer coverage + §11.4.43 TDD-fix discipline, every PASS carrying §11.4.5 captured-evidence (REAL PROOFS, not metadata-only); (5) only then push, verifying with `git ls-remote post-push`. No escape hatch — no `--skip-fetch`, `--no-investigate`, `--fast-push`, `--trust-upstream` flag.

Cascade requirement: This anchor (verbatim or by §11.4.71 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the push-discipline layer. **Canonical authority:** constitution submodule Constitution.md §11.4.71 for the full mandate.

§11.4.72 — Audio Top-Priority Mandate (cascaded from constitution submodule §11.4.72)

Verbatim user mandate (2026-05-20): *"Make sure all fixes for audio are always top priority in main working stream!"*

The conductor (main working stream — Claude Code session, AI agent, or human operator) **MUST** treat audio fixes as the highest-priority class on the serial dispatch queue. Any time the conductor faces a choice between dispatching an audio task vs a non-audio task on the SAME serial resource, the audio task wins. Parallel BACKGROUND subagents (research, refactors, infrastructure documentation) **MAY** run concurrently with audio work but do **NOT** preempt audio on the main-stream serial dispatch queue. No escape hatch — there is no "but this non-audio task is faster" or "but this research is more interesting" override; audio-stack regressions are user-perceptible and high-impact while research and refactors can wait.

Cascade requirement: This anchor (verbatim or by §11.4.72 reference) **MUST** appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a process violation at the dispatch-priority layer. **Canonical authority:** constitution submodule Constitution.md §11.4.72 for the full mandate.

§11.4.73 — Main-Specification Document Versioning + Revision Discipline (cascaded from constitution submodule §11.4.73)

Verbatim user mandate (2026-05-20): *"Make sure everything we add now in previous and upcoming requests IS ALWAYS applied to the main specification — if we have one. Since all these are not major changes we could increase Specification version per change for secondary version instead of the primary. Primary version MUST BE increased for much bigger levels of changes! Add this into root (constitution Submodule) Constitution.md, CLAUDE.md and AGENTS.md as mandatory rule / constraint applicable ONLY IF we have something like the main specification document or we do recognize something like the main specification document. Document MUST BE updated ALWAYS to follow the versioning rules we are applying here + revision and other properties we have!"*

Applies **only when a project recognises a main specification document**. When it does: (1) every additive operator requirement, refinement, or accepted recommendation **MUST** be applied to the spec before or as part of the implementing work; (2) spec versioning has two axes — *primary* (V1/V2/V3, bumped for major rewrites by explicit operator decision, old versions archived) and *secondary* (the §11.4.61 metadata-table Revision integer, bumped for every other change); (3) the metadata table **MUST** stay current (Revision, Last modified, Status summary, Fixed); (4) propag-

ated copies of the rule MUST reference the active specification.V<primary>.md, not a stale archive; (5) on primary bump the old file moves to <spec-dir>/archive/ with Status: superseded. Classification: universal, applicable conditionally per the scope condition.

Cascade requirement: This anchor (verbatim or by §11.4.73 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a release blocker when a project has a main spec and lets it drift.
Canonical authority: constitution submodule Constitution.md §11.4.73 for the full mandate.

§11.4.74 — Submodule-Catalogue-First Discovery + Extend-Don't-Reimplement (cascaded from constitution submodule §11.4.74)

Verbatim user mandate (2026-05-20): *"We MUST ALWAYS check which already developed features / functionalities do exist as a part of our comprehensive Submodules catalogue located in vasic-digital and HelixDevelopment organizations on GitHub and GitLab both! Project MUST BE aware of all its existence so we do not implement same things multiple times if they are already done as some of existing universal, reusable general development purpose Submodules! For any missing features that some Submodules we incorporate may be missing we MUST IMPLEMENT the properly and extend those Submodules further! We do control all of the and we CAN and MUST maintain and extend the regularly! All development cycle rules we have MUST BE applied to them and fully respected!"*

Before scaffolding ANY new module, package, helper, or utility, the contributor (human or AI agent) MUST: (1) survey the canonical Submodule catalogue — vasic-digital and HelixDevelopment on both GitHub AND GitLab; (2) inventory existing Submodules; (3) reuse before reimplement — if a Submodule provides the functionality (or 80%+ of it), add it as a Git submodule rather than write fresh; (4) extend in-place when 80%+ matches but features are missing — add the missing features TO THAT SUBMODULE (PR upstream + bump pointer), never as a duplicating consuming-project helper; (5) apply all development-cycle rules to those Submodules; (6) document the survey result in the feature's tracker entry with a Catalogue-Check: field (reuse <org/repo>@<sha> / extend <org/repo>@<sha> / no-match <date>). Classification: universal.

Cascade requirement: This anchor (verbatim or by §11.4.74 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a process violation; duplicate implementations landed without catalogue check are release blockers. **Canonical authority:** constitution submodule Constitution.md §11.4.74 for the full mandate.